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VEHICLE

Abstract

A vehicle includes a vehicle body, a power storage device, and a cover member. The power storage device includes a power storage module and a lower case. The lower case includes a first protrusion that protrudes downward toward the cover member, the lower case being configured to support the power storage module. The cover member includes a second protrusion that protrudes upward toward the lower case.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-034053 filed on Mar. 6, 2024. The disclosure of the above-identified application, including the specification, drawings, and claims, is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a vehicle.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2021-138228 (JP 2021-138228 A) discloses a vehicle equipped with a battery case. The vehicle includes a protective member disposed outside the battery case in a vehicle width direction.

SUMMARY

[0004] In JP 2021-138228 A, the protective member mitigates an impact from a side to the battery case, but there is room for improvement in mitigating an impact from below.

[0005] The present disclosure provides a vehicle capable of mitigating an impact from below.

[0006] A vehicle according to a first aspect of the present disclosure includes a vehicle body, a power storage device, and a cover member. The power storage device includes a power storage module and a lower case, the power storage device being fixed to a bottom portion of the vehicle body. The lower case includes at least one first protrusion that protrudes downward toward the cover member, the lower case being configured to support the power storage module. The cover member includes at least one second protrusion that protrudes upward toward the lower case, the cover member being configured to cover the power storage device from below and cover the lower case from below.

[0007] In the vehicle according to the first aspect of the present disclosure, as described above, the lower case includes at least one first protrusion that protrudes downward toward the cover member, and the cover member includes at least one second protrusion that protrudes upward toward the lower case. As a result, both the first protrusion and the second protrusion function as a buffer material from below, so that an impact from below on the power storage device can be effectively mitigated.

[0008] In the vehicle according to the first aspect of the present disclosure, the at least one first protrusion may include a plurality of first protrusions disposed side by side with a first space interposed between the first protrusions, and the at least one second protrusion may protrude at a position overlapping the first space in an up-down direction. With such a configuration, interference between the first protrusions and the second protrusion can be suppressed.

[0009] In the vehicle according to the first aspect of the present disclosure, the at least one second protrusion may include a plurality of second protrusions disposed side by side with a second space interposed between the second protrusions, and the at least one first protrusion may protrude at a position overlapping the second space in an up-down direction. With such a configuration, interference between the second protrusions and the first protrusion can be suppressed.

[0010] In the vehicle according to the first aspect of the present disclosure, the at least one first protrusion may be fixed to the lower case, and the at least one second protrusion may be fixed to the cover member. With such a configuration, rigidity of the power storage device can be improved as compared with a case where the first protrusion and the lower case are separated from each other and the second protrusion and the cover member are separated from each other.

[0011] The vehicle according to the first aspect of the present disclosure may further include a first buffer member provided between the at least one first protrusion and the cover member, and a second buffer member provided between the at least one second protrusion and the lower case. With such a configuration, the first buffer member and the second buffer member can more effectively mitigate the impact from below on the power storage device.

[0012] In the vehicle according to the first aspect of the present disclosure, the first buffer member may be fixed to at least one of the at least one first protrusion and the cover member, and the second buffer member may be fixed to at least one of the at least one second protrusion and the lower case. As a result, the first buffer member can be stably fixed between the first protrusion and the cover member. In addition, the second buffer member can be stably fixed between the second protrusion and the lower case.

[0013] In the vehicle according to the first aspect of the present disclosure, the lower case may include a case body portion to which the at least one first protrusion is fixed, the cover member may include a cover body portion to which the at least one second protrusion is fixed, the at least one first protrusion includes a first flange portion fixed to the case body portion, and a first protruding portion that protrudes downward from the first flange portion toward the cover member, and the at least one second protrusion includes a second flange portion fixed to the cover body portion, and a second protruding portion that protrudes upward from the second flange portion toward the lower case. With such a configuration, the first flange portion is provided, whereby rigidity of the first protrusion can be increased, and the first protrusion can be easily fixed to the case body portion. In addition, the second flange portion is provided, whereby rigidity of the second protrusion can be increased, and the second protrusion can be easily fixed to the cover body portion.

[0014] In the vehicle according to the first aspect of the present disclosure, at least a part of the first flange portion may overlap at least a part of the second flange portion in an up-down direction. With such a configuration, formation of a space (a space in which neither the first protrusion nor the second protrusion is disposed) between the first flange portion and the second flange portion in a direction in which the first protrusion and the second protrusion are arranged can be suppressed. As a result, transmission of the impact from below to the power storage module can be suppressed as compared with a case where the space is formed.

[0015] In the vehicle according to the first aspect of the present disclosure, a lower end of the at least one first protrusion may be positioned below an upper end of the at least one second protrusion. With such a configuration, a height of the power storage device in the up-down direction can be reduced.

[0016] In the vehicle according to the first aspect of the present disclosure, each of the at least one first protrusion and the at least one second protrusion may be provided to extend in a front-rear direction or a right-left direction of the vehicle body. With such a configuration, the first protrusion and the second protrusion can mitigate the impact from below on the power storage device over a relatively wide range of the bottom portion of the vehicle body.

[0017] According to the present disclosure, the impact from below on the power storage device including the lower case and the cover member can be mitigated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Features, advantages, and technical and industrial significance of exemplary embodiments will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0019] FIG. 1 is a schematic diagram showing a configuration of a vehicle and a power storage device according to an embodiment;

[0020] FIG. 2 is a cross-sectional view showing the configuration of the power storage device according to the embodiment;

[0021] FIG. 3 is a perspective view showing a configuration of a protrusion fixed to a lower case according to the embodiment;

[0022] FIG. 4 is a perspective view showing a configuration of a protrusion fixed to a share panel according to the embodiment; and

[0023] FIG. 5 is a partially enlarged view showing a detailed configuration in the vicinity of a protrusion in FIG. 2.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] An embodiment of the present disclosure will be described with reference to the drawings. In the drawings to be referred to below, the same or corresponding members are denoted by the same number.

[0025] An X direction, a Y direction, and a Z direction shown in the present specification are directions orthogonal to each other. For example, the X direction and the Y direction are a front-rear direction and a right-left direction of a vehicle **900** described below, respectively. The Z direction is an up-down (vertical) direction. The X direction and the Y direction may be the right-left direction and the front-rear direction of the vehicle **900**, respectively.

[0026] FIG. 1 is a schematic perspective view showing a configuration of the vehicle **900** equipped with a power storage device **1** in the embodiment of the present disclosure. In FIG. 1, for simplification, members, such as tires, are not illustrated.

[0027] The vehicle **900** includes a vehicle body **910**, the power storage device **1**, and a share panel **230**. The power storage device **1** is fixed to the vehicle body **910**. Specifically, the power storage device **1** is fixed to an underbody **911** of the vehicle body **910**. The underbody **911** includes a right frame **911b** and a left frame **911a**. The power storage device **1** is fixed to the right frame **911b** and the left frame **911a** by, for example, bolt fastening. The vehicle **900** includes, for example, a plug-in hybrid electric vehicle (PHEV), a battery electric vehicle (BEV), or a fuel cell electric vehicle (FCEV). The underbody **911** and the share panel **230** are examples of a “bottom portion” and a “cover member” of the present disclosure, respectively.

[0028] The power storage device **1** is, for example, a device that stores power for driving the vehicle **900**. The power storage device **1** includes a plurality of power storage modules **100**, a case **200**, a plurality of dampers **300** (FIG. 2), and a plurality of dampers **400** (FIG. 2). The power storage modules **100** are housed in the case **200**. The damper **300** and the damper **400** are examples of a “first buffer member” and a “second buffer member” of the present disclosure, respectively.

[0029] The case **200** includes an upper cover **210** and a lower case **220**. The upper cover **210** is provided to cover the power storage modules **100** from a Z1 side. The lower case **220** supports the power storage modules **100** from a Z2 side.

[0030] The share panel **230** is provided below the case **200**. The share panel **230** is provided to cover the case **200** (lower case **220**) from below.

[0031] A support portion **220b** is provided on a side surface **220a** of the lower case **220** on a Y1 side. The support portion **220b** is fixed (fastened) to the left frame **911a**. In addition, although not shown in FIG. 1, a support portion fixed (fastened) to the right frame **911b** is provided on a side surface of the lower case **220** on a Y2 side. As a result, the power storage device **1** is fixed to the underbody **911**. The method of fixing the power storage device **1** to the underbody **911** is not limited to the above example.

[0032] The share panel **230** is provided with a plurality of protrusions **233**. Details of the protrusion **233** will be described below.

[0033] FIG. 2 is a cross-sectional view showing the power storage device **1** in the embodiment of the present disclosure. The power storage modules **100** are stacked in the Z direction. Each of the power storage modules **100** includes a plurality of electrodes (for example, bipolar electrodes) (not shown) stacked in the Z direction and a separator (not shown) disposed between the electrodes. A current collector, a heat conductive material, or the like (not shown) is disposed between the power storage modules **100** that are stacked. Note that Y1 indicates a front side of a paper surface, and Y2 indicates a back side of the paper surface.

[0034] The upper cover **210** (FIG. 1) includes an upper cover body portion **211** and a connection

portion **212**. The connection portion **212** constitutes an end portion (end surface) on the Z2 side of the upper cover **210**. The connection portion **212** is formed, for example, in a ring shape. The upper cover body portion **211** is formed to protrude from an inner peripheral edge of the connection portion **212** toward the Z1 side.

[0035] The lower case **220** (FIG. 1) includes a lower case body portion **221**, a connection portion **222**, a pair of connection portions **223**, and a plurality of (three in FIG. 2) protrusions **224**. The lower case **220** is formed of a steel plate (for example, an iron plate). The protrusions **224** may be formed of a high-tensile steel sheet. The lower case body portion **221** and the protrusion **224** are examples of a “case body portion” and a “first protrusion” of the present disclosure, respectively.

[0036] The connection portion **222** is fastened to the connection portion **212** of the upper cover **210** by a bolt or the like (not shown). As a result, the lower case **220** is fixed to the upper cover **210**. Similarly to the connection portion **212**, the connection portion **222** is formed, for example, in a ring shape.

[0037] The lower case body portion **221** is formed to be recessed from an inner peripheral edge of the connection portion **222** toward the Z2 side. As a result, a housing space in which the power storage modules **100** are housed is formed between the lower case body portion **221** and the upper cover body portion **211**.

[0038] Specifically, the lower case body portion **221** includes a side portion **221a** and a bottom portion **221b**. The side portion **221a** extends from the inner peripheral edge of the connection portion **222** to the Z2 side. The side portion **221a** is provided to surround the power storage modules **100** as viewed from the Z1 side. The bottom portion **221b** is connected to a lower end portion of the side portion **221a**. In other words, the side portion **221a** extends from an outer peripheral edge of the bottom portion **221b** to the Z1 side. The bottom portion **221b** extends to be orthogonal to the Z direction.

[0039] The connection portions **223** are fastened to connection portions **232** (described below) of the share panel **230** by a bolt or the like (not shown). As a result, the share panel **230** is fixed to the lower case **220**. One of the connection portions **223** is provided on an X1 side of the lower case body portion **221** and the other of the connection portions **223** is provided on an X2 side of the lower case body portion **221**.

[0040] Each of the connection portions **223** is positioned below the bottom portion **221b** of the lower case body portion **221**. That is, each of the connection portions **223** constitutes an end portion (end surface) on the Z2 side of the lower case **220**. Each of the connection portions **223** is connected to an outer peripheral edge of the connection portion **222** by a linking portion **225**.

[0041] Each of the protrusions **224** is formed to protrude in a direction of the Z2 side toward the share panel **230** (a bottom portion **231b** described below). Each of the protrusions **224** is fixed to the lower case body portion **221**. Specifically, each of the protrusions **224** is fixed to a lower surface **221c** of the bottom portion **221b** of the lower case body portion **221**. For example, each of the protrusions **224** is fixed to the lower surface **221c** by welding.

[0042] The share panel **230** includes a share panel body portion **231**, a pair of the connection portions **232**, and the protrusions **233** (two in FIG. 2). The share panel body portion **231** connects the connection portions **232** to each other. The share panel **230** is formed of a steel plate (for example, an iron plate). The protrusions **233** may be formed of a high-tensile steel sheet. In addition, the share panel body portion **231** and the protrusion **233** are examples of a “cover body portion” and a “second protrusion” of the present disclosure, respectively.

[0043] The connection portions **232** are fastened to the connection portions **223** of the lower case **220** by a bolt or the like (not shown). As a result, the share panel **230** is fixed to the lower case **220**. One of the connection portions **232** is provided on the X1 side of the share panel body portion **231** and the other of the connection portions **232** is provided on the X2 side of the share panel body portion **231**.

[0044] The share panel body portion **231** is formed to be recessed toward the Z2 side with respect

to the connection portions **232**. As a result, a space in which the protrusions **224** and the protrusions **233** are disposed is formed between the lower case **220** (lower case body portion **221**) and the share panel body portion **231**.

[0045] Specifically, the share panel body portion **231** includes a pair of side portions **231a** and a bottom portion **231b**. One of the side portions **231a** extends from an end portion on the share panel body portion **231** side of the connection portion **232** on the X1 side to the Z2 side. The other of the side portions **231a** extends from an end portion on the share panel body portion **231** side of the connection portion **232** on the X2 side to the Z2 side. The bottom portion **231b** connects lower end portions of the side portions **231a** to each other. The bottom portion **231b** extends to be orthogonal to the Z direction.

[0046] In a power storage device in the related art, a protective member or the like mitigates an impact from a side to the power storage device (power storage module), but there is room for improvement in mitigating an impact from below.

[0047] In the present embodiment, each of the protrusions **233** is formed to protrude in a direction of the Z1 side toward the lower case **220** (bottom portion **221b**). That is, the protrusion **224** and the protrusion **233** of which protrusion directions are opposite to each other are provided between the lower case **220** and the share panel **230**.

[0048] Each of the protrusions **233** is fixed to the share panel body portion **231**. Specifically, each of the protrusions **233** is fixed to an upper surface **231c** of the bottom portion **231b** of the share panel body portion **231**. For example, each of the protrusions **233** is fixed to the upper surface **231c** by welding.

[0049] The protrusions **224** are disposed side by side in the X direction with a space S1 therebetween. The protrusions **233** are disposed side by side in the X direction with a space S2 therebetween. The space S1 and the space S2 are examples of a “first space” and a “second space” of the present disclosure, respectively.

[0050] Each of the protrusions **233** protrudes at a position overlapping the space S1 in the Z direction. Specifically, a protruding portion **233b** (described below) of each of the protrusions **233** enters the space S1 from the Z2 side.

[0051] A central protrusion **224** in the X direction among the three protrusions **224** protrudes at a position overlapping the space S2 in the Z direction. Specifically, a protruding portion **224b** (described below) of the central protrusion **224** enters the space S2 from the Z1 side.

[0052] Each of the dampers **300** is disposed between the protrusion **224** and the share panel **230** (bottom portion **231b**). That is, the damper **300** is disposed between each of the protrusions **224** and the share panel **230** (bottom portion **231b**).

[0053] Each of the dampers **400** is disposed between the protrusion **233** and the lower case **220** (bottom portion **221b**). That is, the damper **400** is disposed between each of the protrusions **233** and the lower case **220** (bottom portion **221b**).

[0054] FIG. 3 is a perspective view showing a detailed configuration of the protrusion **224** and the damper **300**. The protrusion **224** is formed to extend in the Y direction. That is, the protrusion **224** is provided to extend in the right-left direction of the vehicle **900**.

[0055] The protrusion **224** has a hat-shaped cross section taken along the X direction, as shown in FIG. 2. Specifically, the protrusion **224** includes a pair of flange portions **224a** and the protruding portion **224b**. The protruding portion **224b** is integrally (continuously) formed with each of the flange portions **224a**. The flange portion **224a** and the protruding portion **224b** are examples of a “first flange portion” and a “first protruding portion” of the present disclosure, respectively.

[0056] Each of the flange portions **224a** is fixed to the lower surface **221c** of the lower case body portion **221**. Each of the flange portions **224a** has a flat plate shape extending along the lower surface **221c**. That is, each of the flange portions **224a** extends to be orthogonal to the Z direction.

[0057] The protruding portion **224b** protrudes in the direction of the Z2 side toward the share panel **230** (FIG. 1) from the flange portions **224a**. The protruding portion **224b** is disposed between the

flange portions **224a** in the X direction and is connected to each of the flange portions **224a**. Specifically, one of the flange portions **224a** is connected to an end portion on the X1 side of the protruding portion **224b**. The other of the flange portions **224a** is connected to an end portion on the X2 side of the protruding portion **224b**.

[0058] The protruding portion **224b** includes a pair of inclined portions **224c** and a lower end portion **224d**. One of the inclined portions **224c** extends from an end portion on the X2 side of the flange portion **224a** on the X1 side to the Z2 side. The other of the inclined portions **224c** extends from an end portion on the X1 side of the flange portion **224a** on the X2 side to the Z2 side. The lower end portion **224d** is an example of a “lower end of a first protrusion” of the present disclosure.

[0059] The lower end portion **224d** is provided at a lower end of the protruding portion **224b**. Specifically, the lower end portion **224d** connects lower end portions of the inclined portions **224c** to each other.

[0060] The lower end portion **224d** extends to be orthogonal to the Z direction. In addition, each of the inclined portions **224c** extends to intersect with the Z direction without being orthogonal to the Z direction. The lower end portion **224d** is disposed at a position that does not overlap the flange portions **224a** (and the inclined portions **224c**) as viewed from the Z2 side.

[0061] The damper **300** is fixed to the lower end portion **224d** from the Z2 side. That is, the damper **300** is provided between the lower end portion **224d** and the share panel **230** (bottom portion **231b** (FIG. 2)).

[0062] The damper **300** extends in the Y direction along the protrusion **224** (lower end portion **224d**). A width W1 of the damper **300** in the X direction is smaller than a width W2 of the lower end portion **224d** in the X direction.

[0063] A width W3 of the protruding portion **224b** in the X direction gradually decreases as the protruding portion **224b** extends to the Z2 side. A cross section (FIG. 2) of the protruding portion **224b** along the X direction has a trapezoidal shape. A minimum value of the width W3 is equal to the width W2 of the lower end portion **224d**.

[0064] FIG. 4 is a perspective view showing a detailed configuration of the protrusion **233** and the damper **400**. The protrusion **233** has the same shape (configuration) as the protrusion **224**. The protrusion **233** is formed to extend in the Y direction. That is, the protrusion **233** is provided to extend in the right-left direction of the vehicle **900**.

[0065] The protrusion **233** includes a pair of flange portions **233a** and a protruding portion **233b**. The protruding portion **233b** is integrally (continuously) formed with each of the flange portions **233a**. The flange portion **233a** and the protruding portion **233b** are examples of a “second flange portion” and a “second protruding portion” of the present disclosure, respectively.

[0066] Each of the flange portions **233a** is fixed to the upper surface **231c** of the share panel body portion **231**. Each of the flange portions **233a** has a flat plate shape extending along the lower surface **221c**. That is, each of the flange portions **233a** extends to be orthogonal to the Z direction.

[0067] The protruding portion **233b** protrudes in the direction of the Z1 side toward the lower case **220** (FIG. 1) from the flange portions **233a**. The protruding portion **233b** is disposed between the flange portions **233a** in the X direction and is connected to each of the flange portions **233a**.

Specifically, one of the flange portions **233a** is connected to an end portion on the X1 side of the protruding portion **233b**. The other of the flange portions **233a** is connected to an end portion on the X2 side of the protruding portion **233b**.

[0068] The protruding portion **233b** includes a pair of inclined portions **233c** and an upper end portion **233d**. One of the inclined portions **233c** extends from an end portion on the X2 side of the flange portion **233a** on the X1 side to the Z1 side. The other of the inclined portions **233c** extends from an end portion on the X1 side of the flange portion **233a** on the X2 side to the Z1 side. The upper end portion **233d** is an example of an “upper end of a second protrusion” of the present disclosure.

[0069] The upper end portion **233d** is provided at an upper end of the protruding portion **233b**. Specifically, the upper end portion **233d** connects upper end portions of the inclined portions **233c** to each other.

[0070] The upper end portion **233d** extends to be orthogonal to the Z direction. In addition, each of the inclined portions **233c** extends to intersect with the Z direction without being orthogonal to the Z direction. The upper end portion **233d** is disposed at a position that does not overlap the flange portions **233a** (and the inclined portions **233c**) as viewed from the Z1 side.

[0071] The damper **400** is fixed to the upper end portion **233d** from the Z1 side. That is, the damper **400** is provided between the upper end portion **233d** and the lower case **220** (bottom portion **221b**).

[0072] The damper **400** extends in the Y direction along the protrusion **233** (upper end portion **233d**). A width W4 of the damper **400** in the X direction is smaller than a width W5 of the upper end portion **233d** in the X direction.

[0073] A width W6 of the protruding portion **233b** in the X direction gradually decreases as the protruding portion **233b** extends to the Z1 side. A cross section (FIG. 2) of the protruding portion **233b** along the X direction has a trapezoidal shape. A minimum value of the width W6 is equal to the width W5 of the upper end portion **233d**.

[0074] FIG. 5 is a partially enlarged view of FIG. 2. The damper **300** includes a buffer layer **310**, an adhesive layer **320**, and an adhesive layer **330**.

[0075] The buffer layer **310** is formed of a foaming material, such as a sponge. Hardness of the damper **300** can be adjusted by adjusting a foaming rate in the buffer layer **310**. For example, when it is significant to prevent an abnormal sound from the damper **300**, the buffer layer **310** is formed to be relatively soft. In addition, when it is significant to improve rigidity via the damper **300**, the buffer layer **310** is formed to be relatively rigid.

[0076] The adhesive layer **320** is formed on an upper surface of the buffer layer **310**. As a result, the damper **300** (buffer layer **310**) is fixed to the protrusion **224** (lower end portion **224d**) by the adhesive layer **320**.

[0077] The adhesive layer **330** is formed on a lower surface of the buffer layer **310**. As a result, the damper **300** (buffer layer **310**) is fixed to the share panel **230** by the adhesive layer **330**. Therefore, the protrusion **224** (lower end portion **224d**) is fixed to the share panel **230**.

[0078] Each of the adhesive layer **320** and the adhesive layer **330** may be formed of an adhesive material, or may be formed of an adhesive tape (double-sided tape). In addition, in order to remove the share panel **230** from the lower case **220** during maintenance or the like, at least one of the adhesive layer **320** and the adhesive layer **330** may be configured to allow for detachment of the adhesive. For example, at least one of the adhesive layer **320** and the adhesive layer **330** may be formed to be relatively thin.

[0079] The damper **400** has the same configuration as the damper **300**. The damper **400** has a buffer layer **410**, an adhesive layer **420**, and an adhesive layer **430**.

[0080] The buffer layer **410** is formed of a foaming material, such as a sponge. The adhesive layer **420** is formed on a lower surface of the buffer layer **410**. As a result, the damper **400** (buffer layer **410**) is fixed to the protrusion **233** (upper end portion **233d**) by the adhesive layer **420**.

[0081] The adhesive layer **430** is formed on an upper surface of the buffer layer **410**. As a result, the damper **400** (buffer layer **410**) is fixed to the lower case **220** by the adhesive layer **430**.

Therefore, the protrusion **233** (upper end portion **233d**) is fixed to the lower case **220**. In addition, in order to remove the share panel **230** from the lower case **220** during maintenance or the like, at least one of the adhesive layer **420** and the adhesive layer **430** may be configured to allow for detachment of the adhesive. For example, at least one of the adhesive layer **420** and the adhesive layer **430** may be formed to be relatively thin.

[0082] The protrusion **224** has a height h1 in the Z direction. The damper **300** has a thickness t1 (for example, several mm) in the Z direction. The height h1 is greater than the thickness t1. The sum of the height h1 and the thickness t1 is equal to a distance D1 between the bottom portion

221b of the lower case **220** and the bottom portion **231b** of the share panel **230**.

[0083] The protrusion **233** has a height **h2** in the Z direction. The damper **400** has a thickness **t2** (for example, several mm) in the Z direction. The height **h2** is greater than the thickness **t2**. The height **h2** is equal to the height **h1** of the protrusion **224**. The thickness **t2** is equal to the thickness **t1** of the damper **300**.

[0084] The lower end portion **224d** of the protrusion **224** is positioned on the **Z2** side with respect to the upper end portion **233d** of the protrusion **233**. Specifically, the lower end portion **224d** (a lower surface of the lower end portion **224d**) and the upper end portion **233d** (an upper surface of the upper end portion **233d**) are separated from each other by a distance **D2** in the Z direction. The distance **D2** is larger than half the distance **D1** between the bottom portion **221b** of the lower case **220** and the bottom portion **231b** of the share panel **230**.

[0085] A part of the flange portion **224a** of the protrusion **224** overlaps a part of the flange portion **233a** of the protrusion **233** in the Z direction. The flange portion **224a** includes an overlapping portion **224e** that overlaps the flange portion **233a** in the Z direction.

[0086] In the protrusion **224** interposed between the protrusions **233** in the X direction, the overlapping portion **224e** is provided in each of the flange portions **224a**. Meanwhile, in the other protrusions **224**, the overlapping portion **224e** is provided solely on the flange portion **224a** on one side (protrusion **233** side) in the X direction.

[0087] In addition, the flange portion **233a** includes an overlapping portion **233c** that overlaps the flange portion **224a** in the Z direction. The overlapping portion **233e** is provided in each of the flange portions **233a**.

[0088] Each of the overlapping portion **224e** and the overlapping portion **233e** has a width **W11** in the X direction. The width **W11** is larger than half a width **W12** of the flange portion **224a** in the X direction (a width **W13** of the flange portion **233a** in the X direction). The width **W12** and the width **W13** are equal to each other.

[0089] Each of the flange portions **224a** does not overlap the protruding portion **233b** (FIG. 4) of the protrusion **233** in the Z direction. As a result, it is possible to suppress excessive reduction in gap between the protrusion **224** and the protrusion **233** in the Z direction due to the flange portion **224a** and the protruding portion **233b** overlapping in the Z direction.

[0090] Each of the flange portions **233a** does not overlap the protruding portion **224b** (FIG. 3) of the protrusion **224** in the Z direction. As a result, it is possible to suppress excessive reduction in gap between the protrusion **224** and the protrusion **233** in the Z direction due to the flange portion **233a** and the protruding portion **224b** overlapping in the Z direction.

[0091] As can be seen from the above description, the protruding portion **224b** (FIG. 3) of the protrusion **224** and the protruding portion **233b** (FIG. 4) of the protrusion **233** do not overlap each other in the Z direction, and are shifted in position from each other in the X direction.

[0092] As described above, in the present embodiment, the lower case **220** includes the protrusion **224** that protrudes in the direction of the **Z2** side toward the share panel **230**. The share panel **230** includes the protrusion **233** that protrudes in the direction of the **Z1** side toward the lower case **220**. As a result, each of the protrusion **224** and the protrusion **233** functions as a buffer member against the impact from below, so that the impact from below can be effectively mitigated as compared with a case where any one of the protrusion **224** and the protrusion **233** is not provided. As a result, it is possible to suppress the transmission of the impact to the power storage module **100**.

[0093] In the embodiment, an example has been described in which the protrusion **233** is disposed at a position overlapping, in the Z direction, the space **S1** between the protrusions **224** arranged in the X direction, but the present disclosure is not limited thereto. The protrusion **224** and the protrusion **233** may be disposed in the same region as viewed in the Z direction.

[0094] In the embodiment, an example has been described in which the protrusions **224** and the protrusions **233** are provided, but the present disclosure is not limited thereto. The number of either or both of the protrusions **224** and the protrusions **233** may be one.

[0095] In the embodiment, an example has been described in which the damper **300** is adhered (fixed) to each of the protrusion **224** and the share panel **230**, but the present disclosure is not limited thereto. The damper **300** may be adhered (fixed) to solely one of the protrusion **224** and the share panel **230**. Similarly, the damper **400** may be adhered (fixed) to solely one of the protrusion **233** and the lower case **220**.

[0096] In the embodiment, an example has been described in which the protrusion **224** is fixed to the share panel **230** by the damper **300**, but the present disclosure is not limited thereto. The protrusion **224** may be directly fixed to the share panel **230** without using the damper **300**. For example, the protrusion **224** may be directly joined to the share panel **230** by an adhesive material, or the protrusion **224** may be welded to the share panel **230**. Similarly, the protrusion **233** may be directly fixed to the lower case **220** without using the damper **400**.

[0097] In the embodiment, an example has been described in which the flange portions **224a** are provided in a pair on each of the X1 side and the X2 side of the protruding portion **224b**, but the present disclosure is not limited thereto. The protrusion provided on the lower case **220** may include a protruding portion and an annular flange portion provided to surround the protruding portion. In addition, the protrusion provided on the share panel **230** side may be configured in the same manner.

[0098] In the embodiment, an example has been described in which a part of the flange portion **224a** and a part of the flange portion **233a** overlap in the Z direction, but the present disclosure is not limited thereto. Any one of the flange portion **224a** and the flange portion **233a** may entirely overlap the other of the flange portion **224a** and the flange portion **233a** in the Z direction.

[0099] An example has been described in which each of the protrusion **224** and the protrusion **233** extends in the right-left direction of the vehicle **900**, the present disclosure is not limited thereto. Each of the protrusion **224** and the protrusion **233** may extend in the front-rear direction of the vehicle **900**. In addition, the direction in which the protrusion **233** extends and the direction in which the protrusion **224** extends may intersect with each other.

[0100] In addition, at least one of the protrusion **224** and the protrusion **233** may be divided into a plurality of parts in the direction in which the at least one of the protrusion **224** and the protrusion **233** extends. In this case, the damper **300** (**400**) may be similarly divided into a plurality of parts. In addition, a plurality of dampers may be provided side by side on a single protrusion. In this case, the dampers are preferably provided at least at both ends and the center of the protrusion in the direction in which the protrusion extends.

[0101] In the embodiment, an example has been described in which the height **h1** of the protrusion **224** and the height **h2** of the protrusion **233** are equal to each other, but the present disclosure is not limited thereto. The height **h1** and the height **h2** may be different from each other.

[0102] In the embodiment, an example has been described in which the protrusion **224** is a separate member from the lower case body portion **221**, but the present disclosure is not limited thereto. The protrusion **224** may be integrally formed with the lower case body portion **221**. For example, the lower case body portion may be deformed to form the protrusion. In addition, the protrusion **233** may be integrally formed with the share panel body portion **231** in the same manner. For example, the share panel body portion may be deformed to form the protrusion.

[0103] In the embodiment, an example has been described in which the cross section of each of the protrusion **224** and the protrusion **233** along the X direction has a hat shape, but the present disclosure is not limited thereto. For example, the cross section of at least one of the protrusion **224** and the protrusion **233** along the X direction may have a rectangular shape, and a recess may be formed in a part of the protruding portion.

[0104] In the embodiment, an example has been described in which each of the flange portions **224a** does not overlap the protruding portion **233b** of the protrusion **233** in the Z direction, but the present disclosure is not limited thereto. The flange portion **224a** may overlap the protruding portion **233b** in the Z direction. In addition, similarly, the flange portion **233a** may overlap the

protruding portion **224b** of the protrusion **224** in the Z direction.

[0105] The configurations of the embodiment and the various modification examples may be combined with each other.

[0106] The embodiment disclosed herein is merely illustrative and not restrictive in all respects. The scope of the present disclosure is defined by the terms of the claims, rather than the above description of the embodiment, and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

Claims

1. A vehicle comprising: a vehicle body; a power storage device; and a cover member, wherein: the power storage device includes a power storage module and a lower case, the power storage device being fixed to a bottom portion of the vehicle body; the lower case includes at least one first protrusion that protrudes downward toward the cover member, the lower case being configured to support the power storage module; and the cover member includes at least one second protrusion that protrudes upward toward the lower case, the cover member being configured to cover the power storage device from below and cover the lower case from below.
 2. The vehicle according to claim 1, wherein: the at least one first protrusion includes a plurality of first protrusions disposed side by side with a first space interposed between the first protrusions; and the at least one second protrusion protrudes at a position overlapping the first space in an up-down direction.
 3. The vehicle according to claim 1, wherein: the at least one second protrusion includes a plurality of second protrusions disposed side by side with a second space interposed between the second protrusions; and the at least one first protrusion protrudes at a position overlapping the second space in an up-down direction.
 4. The vehicle according to claim 1, wherein: the at least one first protrusion is fixed to the lower case; and the at least one second protrusion is fixed to the cover member.
 5. The vehicle according to claim 1, further comprising: a first buffer member provided between the at least one first protrusion and the cover member; and a second buffer member provided between the at least one second protrusion and the lower case.
 6. The vehicle according to claim 5, wherein: the first buffer member is fixed to at least one of the at least one first protrusion and the cover member; and the second buffer member is fixed to at least one of the at least one second protrusion and the lower case.
 7. The vehicle according to claim 1, wherein: the lower case includes a case body portion to which the at least one first protrusion is fixed; the cover member includes a cover body portion to which the at least one second protrusion is fixed; the at least one first protrusion includes a first flange portion fixed to the case body portion, and a first protruding portion that protrudes downward from the first flange portion toward the cover member; and the at least one second protrusion includes a second flange portion fixed to the cover body portion, and a second protruding portion that protrudes upward from the second flange portion toward the lower case.
 8. The vehicle according to claim 7, wherein at least a part of the first flange portion overlaps at least a part of the second flange portion in an up-down direction.
 9. The vehicle according to claim 1, wherein a lower end of the at least one first protrusion is positioned below an upper end of the at least one second protrusion.
 10. The vehicle according to claim 1, wherein each of the at least one first protrusion and the at least one second protrusion is provided to extend in a front-rear direction or a right-left direction of the vehicle body.
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